6.4.7.2 Mirage and looming

The refractive index of air increases with its density. In hot places, air near the ground is hotter than air at a height. Hot air is less dense. Hence, in still air the refractive index of air increases with height. Because of this, the light from tall objects like a tree, try to pass through a medium whose refractive index decreases towards the ground. Hence, the ray of light successively deviates away from the normal at different layers of air and undergoes total internal reflection when the angle of incidence near the ground exceeds the critical angle. This gives an illusion as if the light comes from somewhere below the ground. Because of the shaky nature of the layers of air, the observer feels as if the object is getting reflected by a pool of water (or) wet surface beneath the object as shown in Figure 6.23(a). This phenomenon is called ***mirage***.

In the cold places the refractive index increases towards the ground because the temperature of air close to the ground is lesser than the temperature at a height above the surface of earth. Thus, the density and refractive index of air close to the ground is greater than for air at a height. In the cold regions like glaciers and frozen lakes, the reverse effect of mirage will happen. Hence, an inverted image is formed little above the surface as shown in Figure 6.23(b). This phenomenon is called ***looming***. It is also called as superior mirage, towering and stooping.

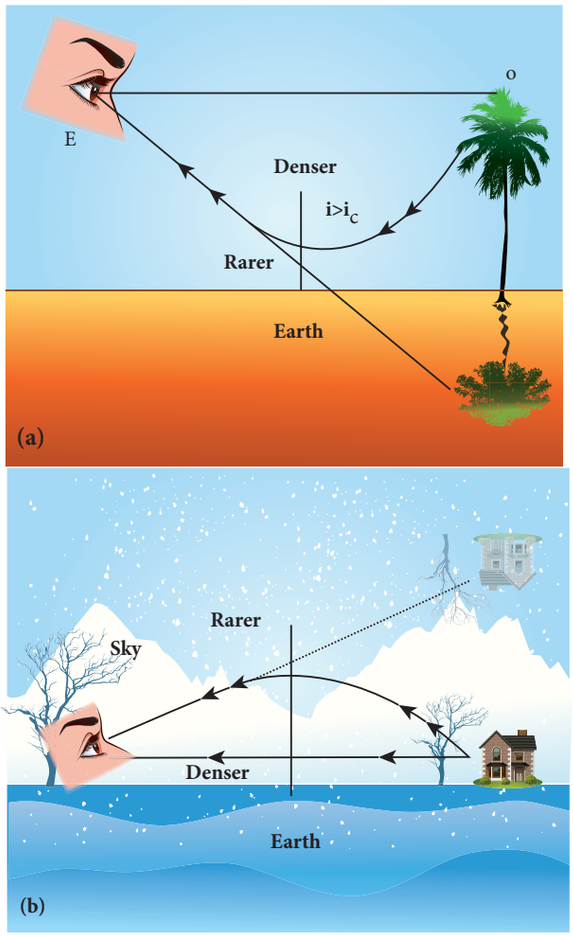


Figure 6.23 (a) Mirage and (b) looming

6.4.7.3 Prisms making using of total internal reflection

Prisms can be designed to reflect light by 90° (or) by 180° by making use of total internal reflection as shown in Figure 6.24(a) and 6.24(b).

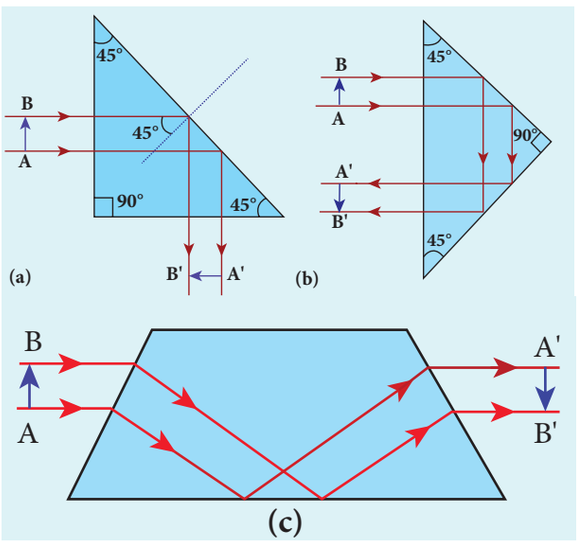


Figure 6.24 Prisms making use of total internal reflection

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